SpaceX Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone | Machine Learning Classification

Data collection • EDA • SQL • Folium • Plotly Dash • Predictive modeling

Executive Summary

$62M vs $165M	90	0% → 84%	83.3%
Falcon 9 price vs. competitors	Falcon 9 launches analysed (83 features)	Landing success, 2010 → 2020	Test accuracy of the classifiers

- **Problem:** SpaceX can reuse the first stage, which is why it can price a launch far below competitors. Predicting whether the stage lands tells us the likely cost of a launch.

- **Approach:** collected launch data (SpaceX API and web scraping), wrangled it, explored it with visualisation and SQL, mapped launch sites with Folium, built a Plotly Dash dashboard, and trained four classifiers.

- **Findings:** success improved steadily as the programme matured; orbit, launch site and payload all matter; Logistic Regression, SVM and KNN each reached **83.3%** test accuracy with identical confusion matrices.

- **Caution:** all errors were false positives (predicting a landing that did not happen), and the test set holds only 18 launches.

Introduction

- **Context:** SpaceX advertises Falcon 9 launches at about **$62 million**; other providers charge upward of **$165 million**. Most of the saving comes from reusing the first stage.

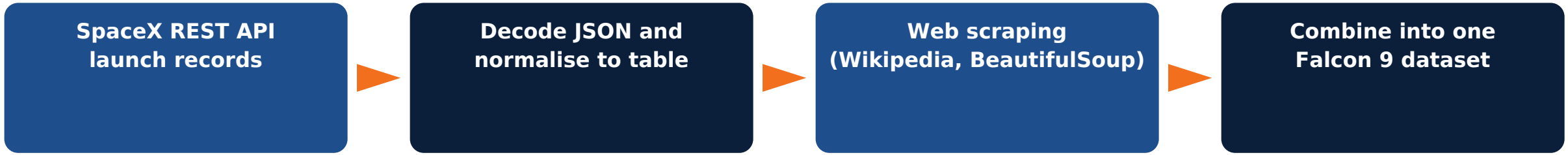
- **Business question:** if an alternative company wants to bid against SpaceX, can it predict whether the first stage will land (and so be reused)?

- **Goal:** build a machine learning pipeline that predicts first-stage landing success from launch characteristics.

Questions we set out to answer

- How has landing success changed over time?
- Which launch sites, orbits and payloads succeed most often?
- Where are the sites, and what is around them?
- Which model predicts landing outcome best?

Methodology: Data Collection



- **SpaceX API:** requested launch, rocket, launchpad, payload and core details; resolved ID fields into readable values (booster version, launch site, payload mass, orbit, outcome, reuse and landing-pad information).

- **Web scraping:** parsed the Falcon 9 launch table from Wikipedia with BeautifulSoup to capture records and fields the API did not provide.

- **Result:** a single dataframe of Falcon 9 launches (flight number, date, booster, payload mass, orbit, launch site, outcome, grid fins, reuse, legs, landing pad, block, serial, coordinates).

Methodology: Data Wrangling

- **Missing values:** replaced missing payload mass with the column mean.
- **Landing outcomes:** counted outcome types (True/False Ocean, RTLS, ASDS, None).
- **Label creation:** mapped each outcome to a binary **Class** column: **1** = first stage landed, **0** = did not land. Planned controlled ocean landings count as successes.
- **Success rate:** averaged Class to get the overall landing rate.

Exploratory Data Analysis

First, let's read the SpaceX dataset into a Pandas dataframe and print its summary

```
]: df=pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_pa

# If you were unable to complete the previous lab correctly you can uncomment and load this csv

# df = pd.read_csv('https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBMDeveloperSkillsNetwork-DS0701EN-SkillsN

df.head(5)
```

]:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004

Wrangled dataset: payload, orbit, launch site, outcome, reuse and landing details

Methodology: EDA and Interactive Visual Analytics

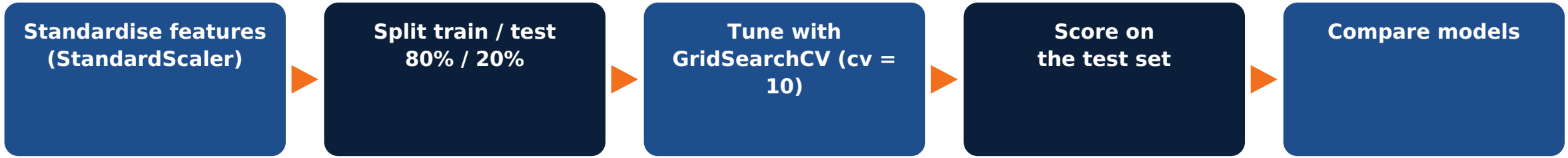
Exploratory analysis

- **Visualisation (Seaborn / Matplotlib):** flight number, payload and orbit versus launch site; success rate by orbit; yearly success trend.
- **SQL (SQLite):** queried the launch table for sites, payload totals, booster versions and landing outcomes.
- **Feature engineering:** extracted the launch year, one-hot encoded categorical columns.

Interactive analytics

- **Folium maps:** marked each launch site, added success/failure markers coloured by Class, grouped them into marker clusters, and measured distances from sites to coastline, railway, highway and city.
- **Plotly Dash:** built a dashboard with a site drop-down, a success pie chart, a payload range slider and a payload-versus-outcome scatter plot coloured by booster version.

Methodology: Predictive Analysis



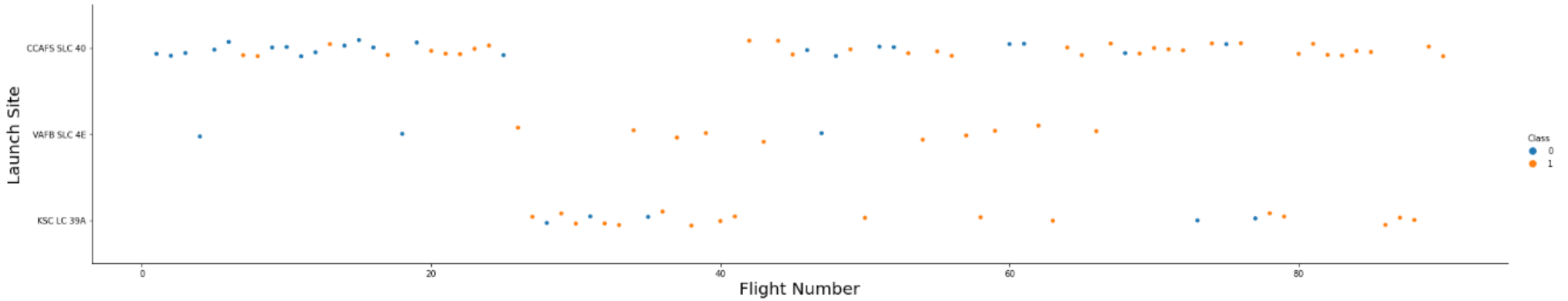
- **Target:** Class (1 = landed, 0 = did not land). **Features:** 83 columns after one-hot encoding.
- **Split:** test_size = 0.2, random_state = 2, giving **72 training** and **18 test** launches.
- **Models and tuned hyperparameters:** Logistic Regression (C, penalty, solver), Support Vector Machine (kernel, C, gamma), Decision Tree (criterion, splitter, depth, features, leaf and split sizes), K-Nearest Neighbors (neighbors, algorithm, p).
- **Selection:** best parameters by 10-fold cross-validated accuracy; final comparison on the held-out test data and confusion matrices.

EDA with Visualization: Flight Number vs. Launch Site

TASK 1: Visualize the relationship between Flight Number and Launch Site

Use the function `catplot` to plot `FlightNumber` vs `LaunchSite`, set the parameter `x` parameter to `FlightNumber`,set the `y` to `Launch Site` and set the parameter `hue` to `'class'`

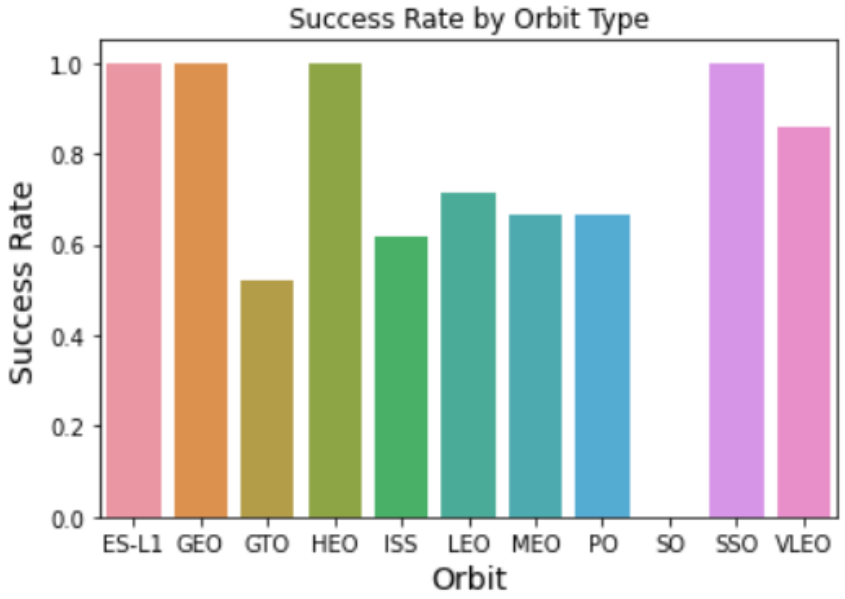
```
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect=5)
plt.xlabel("Flight Number", fontsize=20)
plt.ylabel("Launch Site", fontsize=20)
plt.show()
```



Insight: early flights (blue, Class 0) cluster at CCAFS SLC-40 and VAFB SLC-4E, while later flights (orange, Class 1) are overwhelmingly successful at every site. Success rises with flight number, which points to learning and improvement over time. KSC LC-39A only appears from roughly flight 26 onward.

EDA with Visualization: Success Rate by Orbit

	Orbit	Class
0	ES-L1	1.000000
1	GEO	1.000000
2	GTO	0.518519
3	HEO	1.000000
4	ISS	0.619048
5	LEO	0.714286
6	MEO	0.666667
7	PO	0.666667
8	SO	0.000000
9	SSO	1.000000
10	VLEO	0.857143

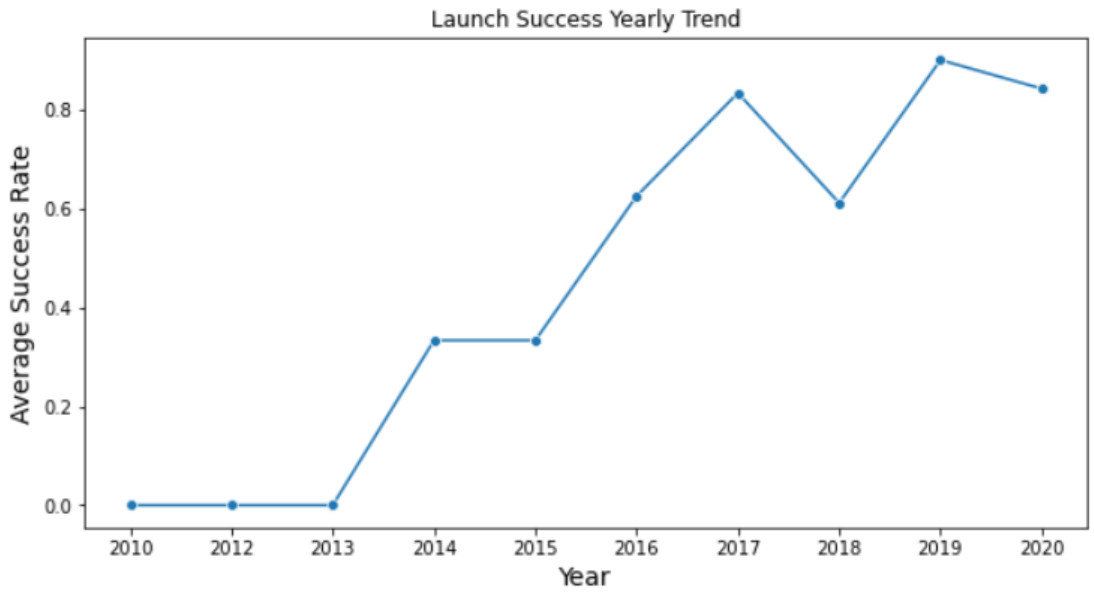


- **100% success:** ES-L1, GEO, HEO and SSO.
- **High:** VLEO (85.7%) and LEO (71.4%).
- **Middle:** MEO and PO (66.7%), ISS (61.9%).
- **Weakest:** GTO (51.9%); SO has 0% success.

Insight: the 100% orbits have very few launches, so their rates are less reliable. GTO is the hardest well-sampled orbit: high-energy missions leave less fuel for recovery, so landing is riskier.

EDA with Visualization: Yearly Launch Success Trend

	Year	Class
0	2010	0.000000
1	2012	0.000000
2	2013	0.000000
3	2014	0.333333
4	2015	0.333333
5	2016	0.625000
6	2017	0.833333
7	2018	0.611111
8	2019	0.900000
9	2020	0.842105



- **2010 to 2013:** 0% landing success.
- **2014 to 2015:** 33%.
- **2016 to 2017:** climbs to 62.5% then 83.3%.
- **2019:** peak of 90%; 2020 holds at 84.2%.

Insight: the trend is strongly upward, with a dip in 2018 (61%) that was recovered. Landing is now routine, which is why reuse can support the lower price.

EDA with SQL: Launch Sites and Records

Display the names of the unique launch sites in the space mission

```
%%sql
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Unique launch sites in the dataset

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

First five launches from sites starting with 'CCA'

Insight: the table lists four site names, but CCAFS LC-40 and CCAFS SLC-40 are the same Cape Canaveral pad written two ways, so SpaceX effectively launches from **three** locations (Cape Canaveral, Kennedy Space Center, Vandenberg). The first 2010 and 2012 launches show parachute failures and "no attempt" landings, before recovery began.

EDA with SQL: Mission and Landing Outcomes

List the total number of successful and failure mission outcomes

```
%%sql
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Mission outcomes (COUNT by Mission_Outcome)

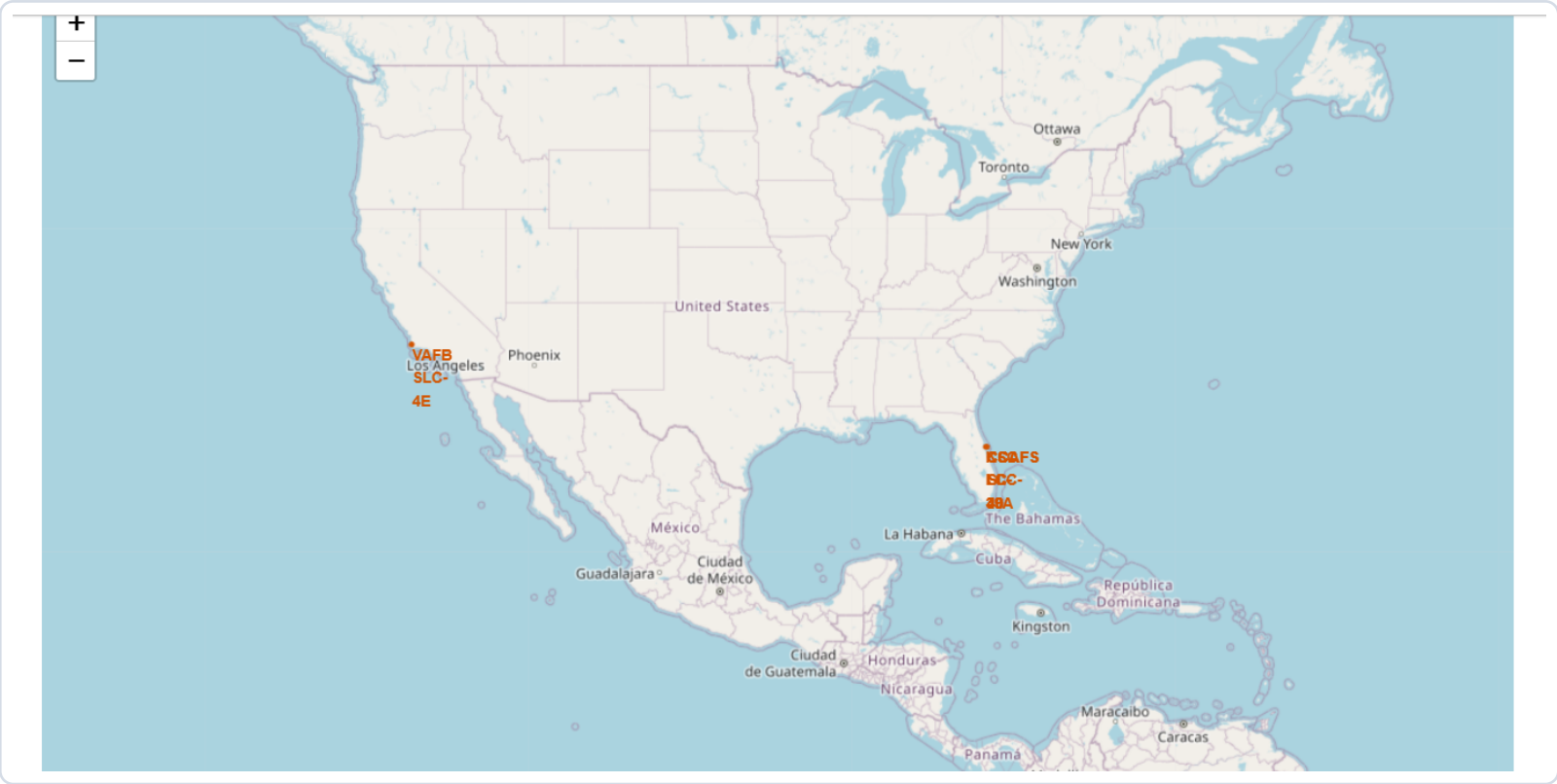
Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Landing outcomes, ranked

- **100 of 101** missions succeeded in delivering the payload.
- Drone-ship landings: 5 succeeded, 5 failed.
- Ground-pad landings: 3 of 3 succeeded.

Insight: mission success (about 99%) is a separate question from landing success. SpaceX delivers payloads reliably even when the booster is lost, so the real uncertainty for a bidder is whether the stage can be recovered. In this ranked window, ground-pad landings were perfect.

Interactive Map (Folium): Launch Site Locations



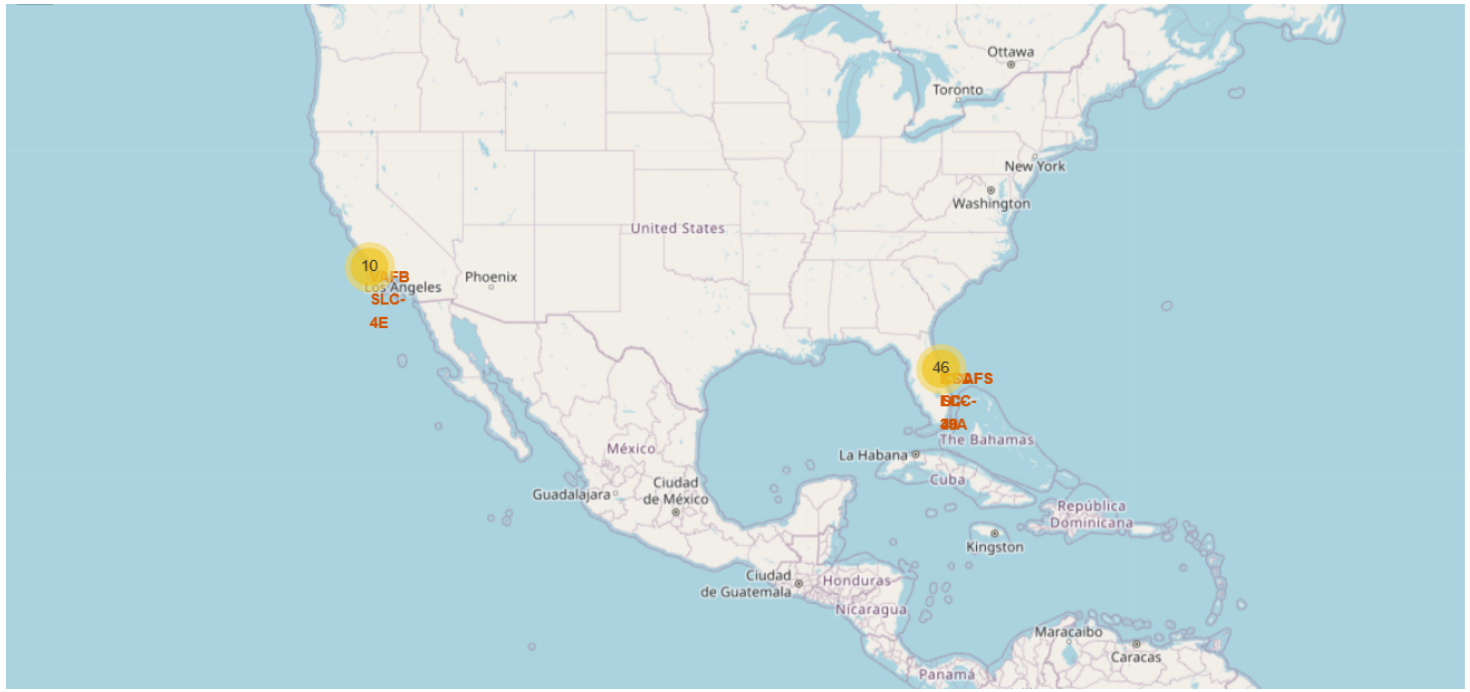
All launch sites marked on an interactive map

	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745

Launch site coordinates

Insight: every site sits right on a coast: three in Florida and one in California. Coastal launches let rockets fly over water, which keeps debris and failed stages away from populated areas.

Interactive Map (Folium): Outcomes and Clusters



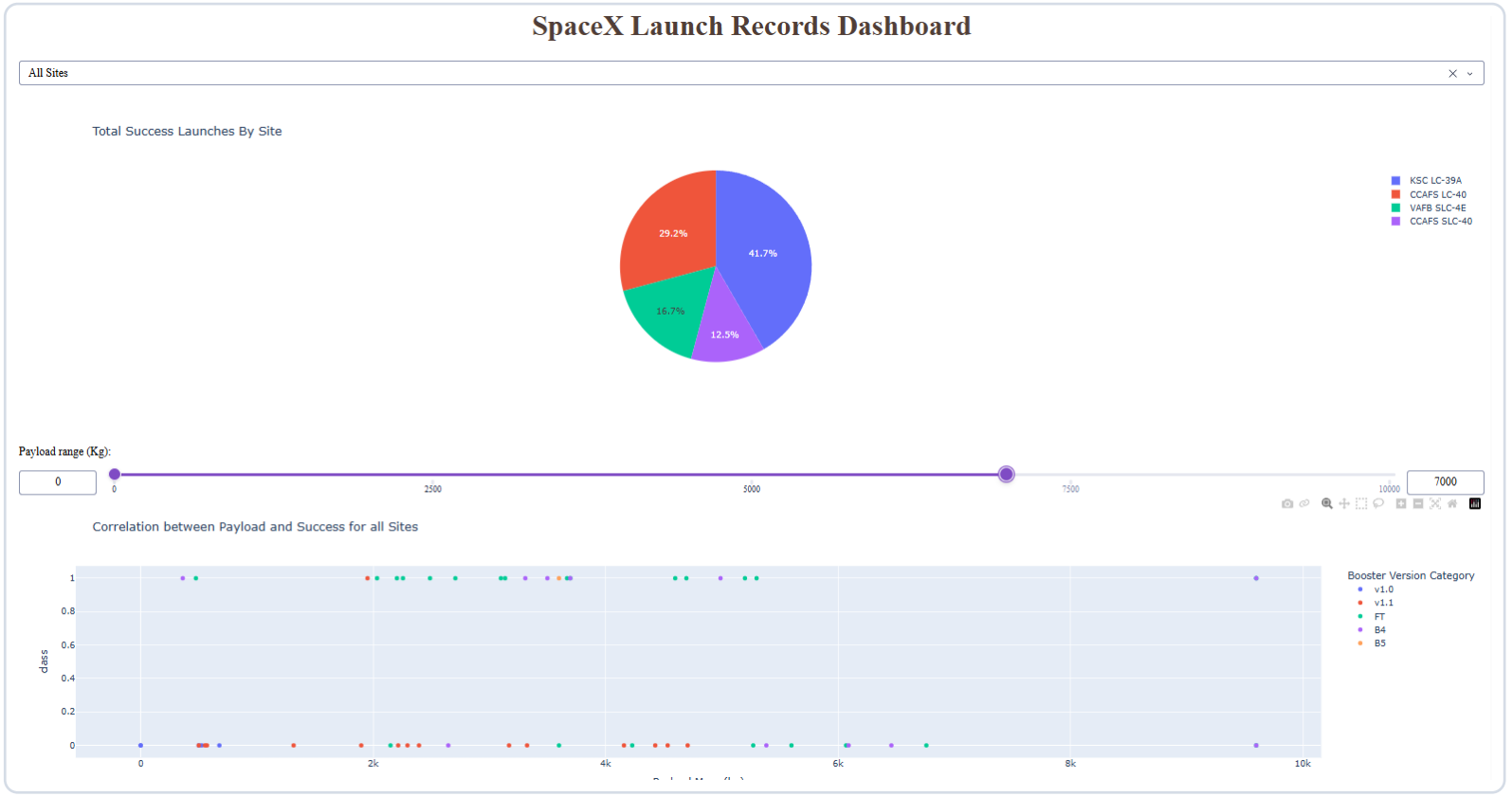
Marker clusters: 46 launches in Florida, 10 in California

	Launch Site	Lat	Long	class
46	KSC LC-39A	28.573255	-80.646895	1
47	KSC LC-39A	28.573255	-80.646895	1
48	KSC LC-39A	28.573255	-80.646895	1
49	CCAFS SLC-40	28.563197	-80.576820	1
50	CCAFS SLC-40	28.563197	-80.576820	1
51	CCAFS SLC-40	28.563197	-80.576820	0
52	CCAFS SLC-40	28.563197	-80.576820	0
53	CCAFS SLC-40	28.563197	-80.576820	0
54	CCAFS SLC-40	28.563197	-80.576820	1
55	CCAFS SLC-40	28.563197	-80.576820	0

Class value per launch

Insight: each marker is one launch with its Class (1 = landed, 0 = failed). Florida hosts 46 of the 56 launches (about 82%), so most of the learning happened there, and zooming in lets us compare outcomes pad by pad.

Plotly Dash Dashboard



- **Site drop-down** filters the whole dashboard.
- **Pie chart:** KSC LC-39A has the largest share of successful launches (41.7%), then CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%).
- **Payload slider** (here 0 to 7,000 kg) updates the scatter plot.

Insight: successes concentrate at mid-range payloads (about 2,000 to 5,500 kg), mostly by FT and B4 boosters. The older v1.0 and v1.1 boosters show mostly failures.

Predictive Analysis: Model Results

`X_train, X_test, Y_train, Y_test`

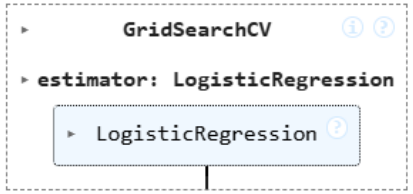
```
X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.2,random_state=2)
print(X_train.shape, X_test.shape, Y_train.shape, Y_test.shape)
```

```
(72, 83) (18, 83) (72,) (18,)
```

Create a logistic regression object then create a GridSearchCV object `logreg_cv` with cv = 10.

```
parameters ={'C':[0.01,0.1,1],
             'penalty':['l2'],
             'solver':['lbfgs']}
```

```
parameters ={"C":[0.01,0.1,1],'penalty':['l2'], 'solver':['lbfgs']}# l1 lasso l2 r
lr=LogisticRegression()
logreg_cv=GridSearchCV(lr,parameters,cv=10)
logreg_cv.fit(X_train,Y_train)
```



GridSearchCV with 10-fold cross-validation

Model	Test accuracy
Logistic Regression	**83.3%**
Support Vector Machine	**83.3%**
Decision Tree	**83.3%**
K-Nearest Neighbors	**83.3%**

Insight: 72 launches trained the models and 18 tested them. All four models reached the same **83.3%** test accuracy (15 of 18 correct), so no single method clearly wins. With such a small test set, one launch changes accuracy by 5.6 points, so cross-validated accuracy is the fairer tiebreaker.

Predictive Analysis: Confusion Matrices

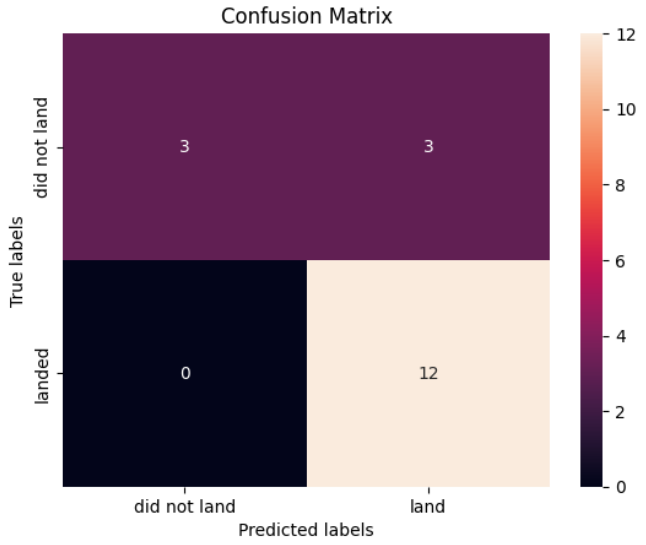
Calculate the accuracy on the test data using the method `score`:

```
print("test accuracy :", logreg_cv.score(X_test, Y_test))
```

```
test accuracy : 0.8333333333333334
```

Lets look at the confusion matrix:

```
yhat=logreg_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



Logistic Regression

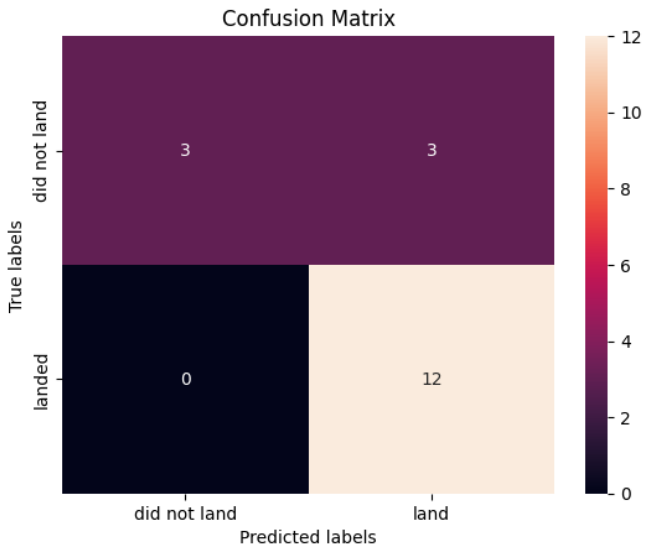
Calculate the accuracy on the test data using the method `score`:

```
print("test accuracy :", svm_cv.score(X_test, Y_test))
```

```
test accuracy : 0.8333333333333334
```

We can plot the confusion matrix

```
yhat=svm_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



Support Vector Machine

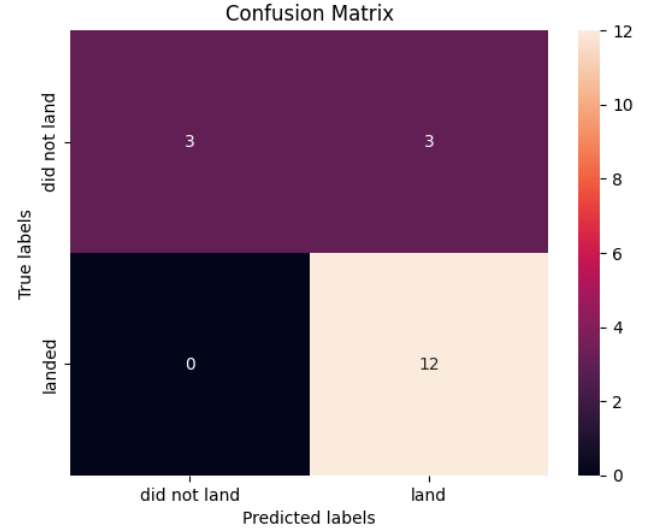
Calculate the accuracy of knn_cv on the test data using the method `score`:

```
print("test accuracy :", knn_cv.score(X_test, Y_test))
```

```
test accuracy : 0.8333333333333334
```

We can plot the confusion matrix

```
yhat = knn_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



K-Nearest Neighbors

Reading the matrix: 12 true positives, 3 true negatives, 3 false positives and 0 false negatives. The models never miss a real landing (100% recall) but wrongly predict a landing in 3 of the 6 failures. The main weakness is being over-optimistic.

Innovative Insights

Cost lens

A reused first stage supports a price near **$62M** against **$165M+**, roughly **$100M** less per launch. A landing prediction becomes a cost estimate for any competing bid.

Optimism risk

All errors were false positives. For a bidder, assuming reuse that does not happen understates cost, so a stricter threshold or class weighting is worth testing.

Maturity effect

Success climbed from 0% to about 84% in a decade. Flight number and year act as a proxy for technology maturity, so recent launches are the best guide to the future.

Site and orbit risk

GTO (52%) and early CCAFS launches were weakest; ground-pad landings were 3 of 3 in the ranked window. Mission type matters as much as the rocket.

Next steps: add more launches (the test set is only 18), compare models with repeated cross-validation, report precision and recall rather than accuracy alone, and add weather and booster-reuse history as new features.

Conclusion

- **Landing success has grown** from 0% (2010 to 2013) to about 84 to 90% (2019 to 2020).
- **Launch site, orbit and payload** all influence the outcome; GTO is hardest, and mid-range payloads with newer boosters succeed most.
- **All launch sites are coastal**: three in Florida and one in California, with most launches from Florida.
- **Four models reached 83.3%** test accuracy, so any of them gives a workable first estimate of landing success.
- **Business value:** predicting landings predicts reuse, and reuse drives the $62M versus $165M price gap.